

TRELU: A NOVEL ACTIVATION FUNCTION FOR MODERN DAY INTRUSION DETECTION SYSTEM USING DEEP NEURAL NETWORKS

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Abstract—The security of computer networks is of paramount importance in today's digital age, where network intrusion attempts are becoming more sophisticated and frequent. Intrusion detection systems (IDS) which were established long ago are used extensively in the industry but are also limited in their ability to detect modern intrusion techniques that are often obfuscated and hidden in network traffic. As a result, there is a growing need for advanced techniques in network security that can effectively identify and mitigate such attacks.

This study proposes a novel activation function TReLU, which is implemented with a hybrid model of CNN and BiLSTM. This function edges out the currently used activation functions for this use case. Additionally, to reduce the training time, parallel programming in Python is leveraged to accelerate the computation of the model. We have used three benchmark IDS datasets to evaluate our proposed model with TReLU. Compared to the previous works of other authors, the experimental results depict that our model outperforms or slightly edges them and various other state of the art models.

Index Terms—TReLU, CNN-BiLSTM, Activation Functions, IDS, Hybrid Model

I. INTRODUCTION

In recent years, network security has grown to be a significant barrier to the advancement of network technologies and the security of computer networks has become a crucial concern due to the increasing frequency and complexity of network infiltration attempts. To tackle the unexpected and rapid growth of network intrusion attempts, intrusion detection is a crucial mechanism that actively defends against attacks, which can increase network security. Traditional intrusion detection systems (IDS) have limitations in identifying contemporary intrusion techniques, which are often disguised and concealed within network traffic. Therefore, there is a growing demand

for more sophisticated IDSs that can quickly and accurately detect and prevent such attacks.

For the research and development of modern-day IDS, there are several benchmark datasets available publicly of which we used the UNSW-NB15, NSL-KDD, and CICIDS2017. While these are used quite extensively for doing research and training purposes in the domain of Intrusion Detection, they inherit some concerning challenges :

- Data Imbalance
- Increased training times for these large datasets
- Overfitting of the model leading to poor F1-score.

In response to these challenges, we are proposing a novel activation function which we have implemented with a deep learning neural network which is a combination of CNN and Bidirectional LSTM [1]. The proposed model uses the core feature extraction abilities of CNNs for spatial feature extraction and BiLSTM networks for temporal (time-based) feature extraction and merges them to accurately identify a wide range of network intrusions, including those that are difficult to detect using conventional IDS techniques which are based on traditional Machine Learning models.

The data imbalance is taken care of by the sampling technique which uses SMOTE to oversample the data. Furthermore, we have leveraged parallel programming in Python, to reduce the training time drastically and accelerate computation to improve the model performance and a custom activation function that enhances the model's ability to detect subtle variations in network traffic.

To evaluate the effectiveness of our methodology, experiments are conducted on the publicly available datasets mentioned before. The experimental results show that the proposed activation function outperforms the majority of the existing functions in terms of the following metrics: accuracy, precision, recall, and F1 score.

II. RELATED WORKS

The use and application of Machine Learning techniques for anomaly detection in intrusion detection is widespread in network security. Researchers have worked extensively on various ML algorithms. They have developed ML models which can be deployed directly to Intrusion Detection Systems in a real-life setup that is used to make distinctions between benign network traffic and attack traffic.

Deep Learning was applied by Tang et al. [7] in their study. They proposed a Deep Neural Network for Intrusion Detection in an SDN environment. Their model achieved good performance on the NSL-KDD dataset. These authors again published a study later on, where they worked on Recurring neural networks (RNNs) in IDS. They have proposed a gating mechanism that is being used in GRU-RNN and has evaluated their model primarily on the NSL-KDD dataset. The authors of this paper [8] propose that there is an improvement in anomaly detection accuracy (89%), as the GRU-RNN doesn't hinder the network performance.

K. Jiang et al. [1] proposed a hybrid model of two Deep Neural Networks - CNN and Bi-LSTM where CNN was used to extract the spatial features and Bi-LSTM was used to extract the temporal features. They evaluated their model primarily on the UNSW-NB15 and NSL-KDD datasets and achieved accuracies of 77.16% and 83.58%, respectively.

Previous research in IDS has explored the use of various activation functions with both CNN and LSTM models. For instance, the rectified linear unit (ReLU) activation function has been widely used with CNNs due to its simplicity and computational efficiency (Nair & Hinton, 2010) [9]. Another common activation function used with CNNs is the exponential linear unit (ELU), which has been shown to improve model performance in certain cases compared to ReLU (Clevert, Unterthiner, & Hochreiter, 2016) [10]. We also have The Bent Identity Activation function which is a non-linear activation function that was introduced by Hafemann et al. [11] This function is a smooth approximation of the ReLU activation function and has a similar shape. The main difference is that the Bent Identity Activation function is differentiable at $x=0$, whereas ReLU is not, which makes it well-suited for gradient-based optimization methods, which are commonly used in deep learning.

In the context of LSTM models, the hyperbolic tangent (tanh) activation function is often used for its ability to squash input values between -1 and 1, making it easier to propagate gradients through time (Hochreiter & Schmidhuber, 1997) [12]. Another popular activation function used with LSTMs is the sigmoid function, which is commonly used for binary

classification tasks (Gers & Schmidhuber, 2001) [13].

Sinha et al. [14] suggested the use of K-fold cross-validation in a CNN BiLSTM network model. They evaluated their model two datasets namely NSL-KDD and UNSW-NB15 datasets. They obtained an accuracy of 82.08% on the UNSW dataset and 99.22% on the NSL-KDD dataset.

During our research we observed that previous papers haven't explored the effects of activation functions for the purpose of overfitting on IDS in great depth, moreover, these functions face some inherent limitations like the dying ReLU, vanishing gradient, unbounded & unsaturated nature of Leaky ReLU in the negative half & the bounded, saturated nature of tanh in the positive half.

III. METHODOLOGY

A. System Architecture

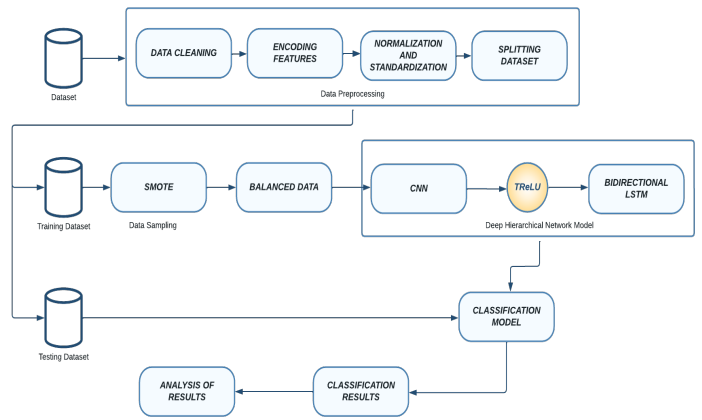


Fig. 1. System Architecture Of Our Proposed System

As shown in Fig. 1, we are using our activation function (TReLU) with the above model architecture. It consists of a SMOTE sampling method and a hybrid model (a combination of CNN and BiLSTM) to obtain both spatial and temporal information. We have applied TReLU after the CNN layer.

The use of TReLU with CNN is because CNN layers are effective in capturing this information by applying convolutional filters. Adding an activation function like the proposed one after a CNN layer can help to further enhance the features extracted by the CNN layer. During the training process, Stratified K Cross Validation is applied to improve the efficiency of the model.

B. Models

1) CNN

CNNs work by first extracting the local features and then for reducing the dimensions, pooling layer downsamples the feature maps and finally the fully connected layers are used to classify the data. CNN in IDS is applied on network traffic data to extract local spatial information. This is important because network traffic data is often highly complex and contains a large amount of noise.

Using this the model observes significant patterns and correlations in the data which are used to detect anomalies that are detected as intrusions.

2) BiLSTM

Long Short-Term Memory networks are RNN based which are used to handle time-series data by incorporating specific gates. By analyzing the local features over time, the LSTM is able to identify patterns and relationships that may not be apparent in the original data. This allows the model to detect anomalies that occur over longer time periods, such as slow attacks or stealthy behavior. BiLSTM models are used so that information that is to be obtained in the future can also be used to train the network's current state.

3) CNN-BiLSTM

In a hybrid CNN-BiLSTM model, the CNN and BiLSTM are combined to take advantage of their complementary strengths. The CNN extracts local features from the data, which are then fed into the LSTM to capture temporal dependencies. This allows the model to effectively analyze complex time-series data and detect anomalous behavior with high accuracy.

C. Activation Functions

Activation functions introduce non-linearity to the network. They transform the output of a neuron into a form that is suitable for the input of the next neuron.

1) ReLU

It stands for Rectified Linear Unit. All negative values in the input are set to zero, and the positive values stay as it is, resulting in a simple and efficient nonlinearity that helps prevent the vanishing gradient problem. However, ReLU can suffer from the "dying ReLU" problem, where neurons become unresponsive if the input is negative. Since this function is linear, its non-saturating and hence results in quick convergence of gradient towards global minima.

2) Leaky ReLU

Leaky ReLU is just an extended version of ReLU that multiplies the negative values with a very small constant (eg 0.01) hence removing the problem of "dying ReLU" and hence improving the performance. It might result in slower predictions for negative values when compared to ReLU as it accounts for them. Although the small slope in the negative region can help prevent the dying ReLU problem, it can still cause some neurons to become unresponsive if the slope is not set correctly.

3) Tanh

Tanh (Hyperbolic Tangent) is a classic activation function that maps the input to a range of -1 to 1. Tanh also has the vanishing gradient problem. It satisfies the property of having zero mean values, which is desirable in the context of neural networks.

4) TReLU

Here we propose a novel hybrid activation function that combines the benefits of both tanh and linear activation

functions. We show that this activation function outperforms other popular activation functions in terms of classification accuracy and convergence speed on three of the most extensively used datasets: UNSW, CICIDS, and NSL-KDD.

We are proposing a novel activation function, which is a hybrid of parameterized tanh and ReLU functions, and it can be written as:

$$f(x) = \begin{cases} \alpha \cdot \tanh(x) & x < 0 \\ x & x \geq 0 \end{cases} \quad (1)$$

where α is a hyperparameter.

Graphical depiction of the function on various values of α is as shown in the Fig. 2.

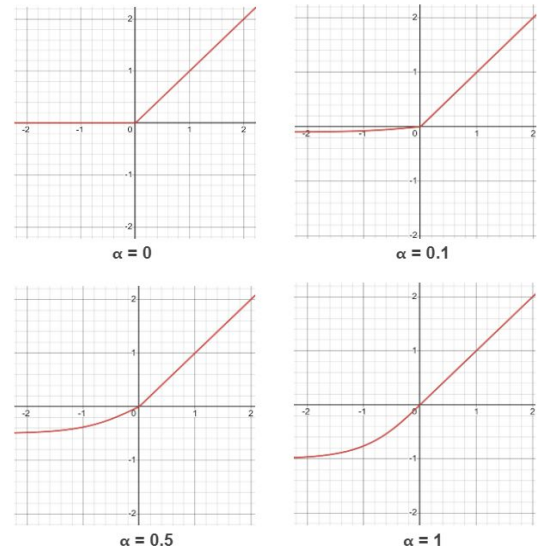


Fig. 2. TReLU function on various values of α

Let x be an input vector of size n , and W be a weight matrix of size $m \times n$. Let b be a bias vector of size m . Then the output of a single layer in a CNN or BiLSTM architecture can be written as:

$$z = W \cdot x + b \quad (2)$$

where z is a vector of size m .

Next, we apply the proposed activation function $f(z)$ to get the final output of the layer:

$$a = f(z) = \begin{cases} \alpha \cdot \tanh(z) & z < 0 \\ z & z \geq 0 \end{cases} \quad (3)$$

The derivative of $f(z)$ with respect to z can be calculated as:

$$f'(z) = \begin{cases} \alpha \cdot \text{sech}^2(z) & z < 0 \\ 1 & z \geq 0 \end{cases} \quad (4)$$

It can be observed that the proposed activation function is differentiable, and the derivative is continuous at all

points.

The use of tanh in the negative half of the function is motivated by its ability to provide non-linearity and saturation, which can be useful for modeling complex decision boundaries. The use of identity in the positive half of the function is motivated by its simplicity and ability to preserve information, which can be essential for preserving important features and avoiding overfitting.

Compared to the ReLU function, the proposed activation function has a smoother transition from the negative to positive values of x . These smoother transition results in faster convergence, effective optimization, and generalization.

This activation function also has a parameter α that controls the slope of the function for negative values of x , allowing for some degree of flexibility in the shape of the activation function. Adding alpha limits the point the graph goes to in the third quadrant. The minimum value of x that can be achieved equals $-\alpha$. This can be proven as follows:

Minimum value of $\tanh(x)$ can be found as follows.

Let $y = \tanh(x)$, then we have:

$$y = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (5)$$

Multiplying the num. and denom. by e^x , we get:

$$y = \frac{e^{2x} - 1}{e^{2x} + 1} \quad (6)$$

Now we need to find the limit of y as x approaches $-\infty$. As x approaches $-\infty$, we have:

$$\lim_{x \rightarrow -\infty} e^{2x} = 0 \quad (7)$$

Thus, we get:

$$\lim_{x \rightarrow -\infty} y = \lim_{x \rightarrow -\infty} \frac{e^{2x} - 1}{e^{2x} + 1} = -1 \quad (8)$$

Since, we know that $\tanh(x)$ is always increasing function, we know that the minimum value of $\tanh(x)$ is -1 . Hence at the minimum value of $\tanh(x)$, the value of our function will be:

$$\lim_{x \rightarrow -\infty} f(x) = \alpha \cdot -1 \quad (9)$$

Therefore from (9), we can conclude that the minimum value of x possible using the proposed activation function is $-\alpha$.

Thus, this can be useful for tuning the neural network to different types of attacks or for dealing with imbalanced datasets.

IV. EXPERIMENT RESULTS AND ANALYSIS

This section presents a detailed overview of the datasets used, data preparation techniques and experimental results and analysis of our proposed model with TReLU.

A. Datasets

1) NSL-KDD

This dataset is a NIDS dataset that was created as an upgrade over its original predecessor, KDD Cup 1999 dataset, which had several limitations and weaknesses. The dataset contains a total of 5 classes, 148517 samples, and 41 features, including packet-level information like protocol type, service, IP address of source and destination, and port numbers. [15]

2) UNSW-NB15

This dataset comprises normal network traffic and attack samples. The Dataset contains a total of 257673 records and 49 (excluding class labels) features including packet-level information such as port no, protocol, and IP addresses, and flow-level information such as the no. of packets and total bytes transferred in a given time period. [16]

3) CICIDS2017

This dataset is a relatively newer one and is used for the evaluation of network intrusions. It consists of benign and modern-day attacks, simulating real-world data. It contains network traffic analysis data, the timestamps, IP addresses of the source and destination, ports, protocols, and attacks. There are 78 feature characteristics and one label attribute in the training data. The dataset includes a wide range of attacks ranging from, Brute Force SSH and FTP, Botnet, Heartbleed, etc. [17]

B. Data Preparation

1) One Hot Encoding

Categorical features are present in the datasets used. We convert these categorical values to numerical values in order to yield good prediction scores as numerical data is easier to train on.

2) Normalization

It is the process of transforming numerical data to a standard scale. We are transforming the numerical data to the range of $[0, 1]$.

3) Stratified K-Cross Validation

This is an extended version of the K-fold cross validation method where we divide the data into K parts and this technique runs for K iterations where on the i th iteration, the i th part is used as a validation set while others are used for training the model. The stratifying technique ensures that every split is a nice depiction of the dataset in its entirety. Additionally, we are parallelizing this process by running the K cross on separate threads. This reduced the time drastically to train the model. [18]

C. Results

1) Comparison of Activation functions: We compared how our proposed activation function pans out against other activation functions commonly used in the literature: ReLU, LeakyReLU, and Tanh. Then we also compare them using standard metrics such as Accuracy (ACC), Precision (PR), Recall (R), and F1-Score (F1) of our model for each dataset

with the models previously proposed by other researchers. We first analyze the performance of the TReLU, our proposed activation function on the NSL-KDD dataset. The observed accuracy using TReLU was 84.57%.

TABLE I
COMPARISON OF CLASSIFICATION RESULTS ACROSS DIFFERENT
ACTIVATION FUNCTIONS ON NSL-KDD.

Activation Functions	Evaluation Metrics			
	ACC(%)	PR(%)	R(%)	F1(%)
ReLU	83.47	85.01	85.42	84.21
Leaky ReLU	68.85	72.79	68.85	70.42
Tanh	80.72	79.29	80.72	78.09
TReLU (Ours)	84.57	85.53	84.58	83.30

Then we evaluated the performance of our proposed activation function with the other activation functions on the UNSW-NB15 dataset. The observed accuracy using TReLU was 85.44%.

TABLE II
COMPARISON OF CLASSIFICATION RESULTS ACROSS DIFFERENT
ACTIVATION FUNCTIONS ON UNSW-NB15.

Activation Functions	Evaluation Metrics			
	ACC(%)	PR(%)	R(%)	F1(%)
ReLU	85.33	84.46	85.33	82.8254
Leaky ReLU	85.24	85.00	85.18	82.81
Tanh	84.65	84.72	84.52	83.21
TReLU (Ours)	85.44	85.92	85.37	83.13

Lastly, evaluation of the performance of TReLU was done for the CICIDS2017 dataset. The observed accuracy using TReLU was 98.41%.

TABLE III
COMPARISON OF CLASSIFICATION RESULTS ACROSS DIFFERENT
ACTIVATION FUNCTIONS ON CICIDS-2017.

Activation Functions	Evaluation Metrics			
	ACC(%)	PR	R	F1
ReLU	97.8	97.88	97.36	97.47
Leaky ReLU	98.1	98.35	98.11	98.16
Tanh	96.7	97.42	97.13	97.21
TReLU (Ours)	98.41	98.48	98.29	98.32

Hence, it can be observed that our proposed activation function either slightly edges out or is at par with the other activation functions being used currently for IDS for most of the cases.

2) **Plotted ROC Curves:** The ROC curve is an evaluation metric for classification models. In this plot, TPR (True Positive Rate) is plotted at varying classification thresholds against FPR. It lies between the two extreme points (0,0) and (1,1). The curve can be summarized by the AUC (Area Under Curve) metric, which ranges from 0 to 1, with a higher value indicating better classification performance.

3) **Multi-Class Comparison with other Classifiers:** We have also compared the performance of our deep neural network comprising CNN and Bidirectional LSTM along with TReLU against different classification models. These values

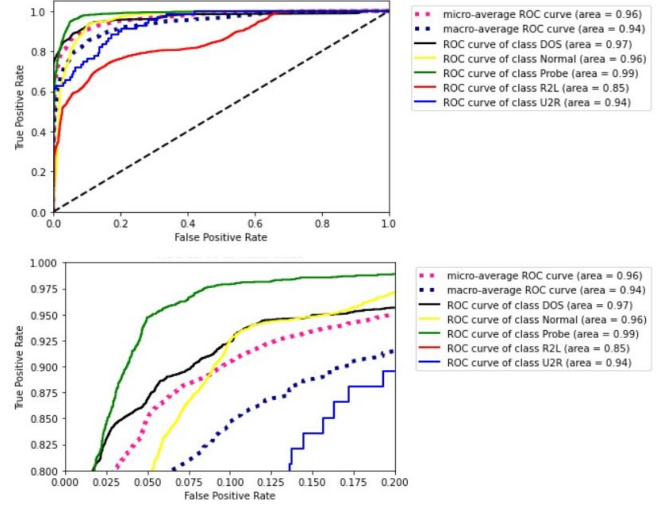


Fig. 3. Multi-class ROC curve of the NSL-KDD dataset

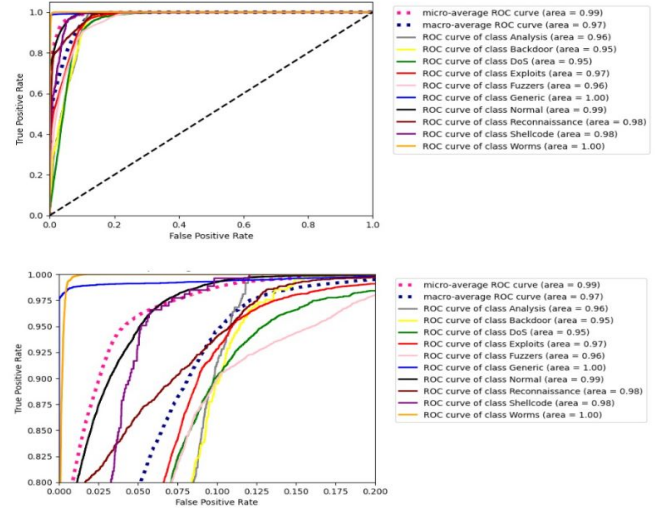


Fig. 4. Multi-class ROC curve of the UNSW-NB15 dataset

have been obtained from various papers as referenced in the tables IV & V.

We have chosen four main metrics for evaluation: weighted accuracy (ACC), weighted precision (PR), weighted recall (R), and the F1-score (F1) for UNSW-NB15 and NSL-KDD datasets. Comparing it with the other models, we can summarise that CNN-BiLSTM with TReLU performs better than the other Machine and Deep Learning models.

For NSL-KDD (Table IV), KDDTest+ is used for evaluating the performance of our model. We can see that our model has a better accuracy over most of the conventional Machine Deep Learning models like SVM, 1D CNN. Only Jiang et al.'s model [1] has a better performance in terms of F1-Measure and Precision. Our model has an accuracy of 84.57% which is 4.44% higher than conventional CNN, and 5.14% higher than conventional BiLSTM. F1-Score for our model is 83.30 which is comparatively much better than most of the other

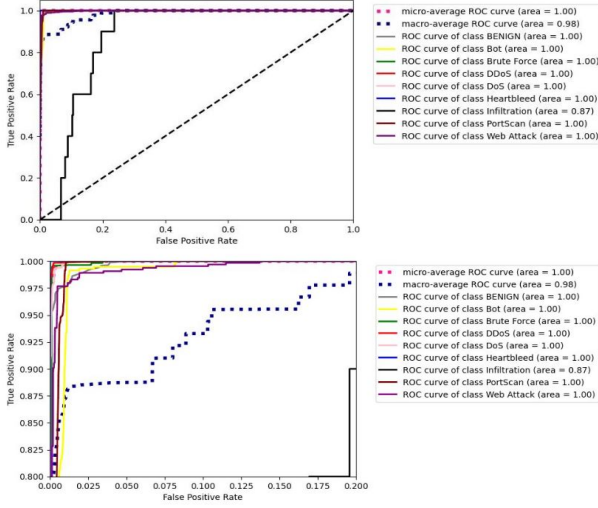


Fig. 5. Multi-class ROC curve of the CICIDS-2017 dataset

classifiers. A better F1-score indicates that the model is able to efficiently classify the samples into its correct class, and hence the False positive rate is low. Generally, F1-scores greater than 0.8 (>80) are considered to be good for any classifier.

TABLE IV
COMPARISON OF MULTICLASS CLASSIFICATION RESULTS ACROSS
DIFFERENT CLASSIFIERS ON NSL-KDD (KDDTEST+).

Model	Evaluation Metrics			
	ACC(%)	PR(%)	R(%)	F1(%)
Naive Bayes [7]	76.56	-	-	-
Random Forest [19]	77.07	82.20	77.10	73.10
SVM [19]	77.12	80.80	77.12	73.90
DNN [7]	75.75	83	76	75
CNN [20]	80.13	-	-	-
RNN [21]	81.29	83.07	81.29	79.25
BiLSTM [1]	79.43	81.14	79.65	80.39
CNN-BiLSTM [1]	83.58	85.82	84.49	85.14
CNN-BiLSTM with TReLU	84.57	85.53	84.58	83.30

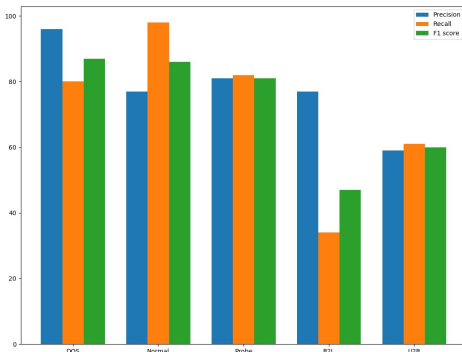


Fig. 6. Classification results of different categories(classes) on NSL-KDD.

To compare the performance of UNSW-NB15 (Table V), UNSW_NB15_testing-set.csv acts as a testing set to evaluate the model. We can see that our classifier has better accuracy

than the other Deep Learning models proposed by other authors. Our model has an accuracy of 85.44% which is 8.28% higher than the CNN-BiDirectionalLSTM model proposed by [1], and edges out the CNN-LSTM hybrid of [21] by 0.3%. We are able to achieve an F1-score of more than 80 (83.13) which indicates that the model's classification is efficient and has a low FPR. Even though it is less than [22]'s F1-score, it still performs better than the rest.

TABLE V
COMPARISON OF MULTICLASS CLASSIFICATION RESULTS ACROSS
DIFFERENT ACTIVATION FUNCTIONS ON UNSW-NB15.

Model	Evaluation Metrics			
	ACC(%)	PR(%)	R(%)	F1(%)
TACGAN-IDS [21]	-	74.91	84.97	79.64
CNN [1]	74.61	81.01	75.65	78.24
CNN-LSTM [22]	85.11	79.57	98.16	87.89
CNN-BiLSTM [1]	77.16	82.63	79.91	81.25
CNN-BiLSTM with TReLU	85.44	85.92	85.37	83.13

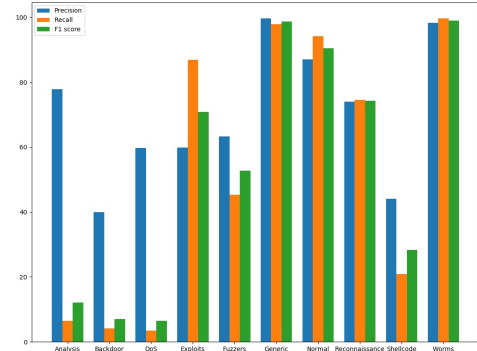


Fig. 7. Classification results of different categories(classes) on UNSW-NB15.

V. CONCLUSION AND FUTURE SCOPE

In this paper, we designed a novel activation function, TReLU(1), for use in IDS with CNN and BiLSTM architectures. We showed through mathematical analysis and experimental evaluation that this function offers advantages over existing activation functions, including improved performance in terms of ACC and F1. We have been able to cut down training time by introducing parallelism in K-folds while training. However, we also acknowledge that further research is needed to grasp the strengths and limitations of the proposed function entirely and to explore its potential applications in other domains.

TReLU has a limitation that it is sensitive to the initialization of the parameter, α i.e. the coefficient of tanh. Thus making α learnable can potentially improve our model. Another promising direction for the next steps is to analyze and explore the performance and efficiency of the proposed function on other datasets and network architectures, to determine its generalizability across different scenarios. The use of Generative Adversarial Networks (GANS) can be done to deal with the issue put forth by high imbalance ratios.

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